

Decomposition of Concentration-Index Trees

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1 4. Results

The results section reports only the objects needed to answer the main technical question: whether inequality-oriented recursive partitioning identifies subgroup structure that is visible, stable, and comparable with classical decomposition output. Full generated outputs are moved to the appendix.

1.1 4.1 Exploratory Data Analysis



Figure 1: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

1.2 4.2 Application of developed tree models to the DRC DHS Data

1.2.1 4.2.1 Published 2007 Validation Benchmark

The generated objects use the DR Congo 2007 analysis data from the previous validated workflow. The table below compares the published DR Congo concentration-index decomposition from Van Malderen et al. (2013, Table 3) with the regression decomposition regenerated in this report. The regenerated model uses the same determinant set as the published survey-GLM comparison, including the household-wealth group quint.

Table 1: Published DR Congo 2007 decomposition benchmark compared with the regenerated regression decomposition

Determinant	Published C contribution (%)	Generated CI contribution (%)	Difference
Household wealth	13.1	9.12	-3.98
Skilled birth attendance	11.3	7.85	-3.45
Sex of the child	-0.6	0.42	1.02
Birth order and interval	-1.2	1.12	2.32
Mother's age at birth	0.4	0.26	-0.14
Type of residence	12.5	8.73	-3.77
Mother's education	33.1	23.06	-10.04
Father's education	-2.1	1.46	3.56
Mother's occupation	17.6	12.28	-5.32
Father's occupation	20.8	21.23	0.43
Region	-4.9	14.47	19.37

1.2.2 4.2.2 Comparison of Indirect Decomposition Results

Categorical predictors enter the regression decomposition as separate one-hot terms. The predictive ranger is fitted on the same design-matrix terms, so the comparison table reports categorical levels at the same granularity instead of collapsing them back to their parent variable.

Table 2: Regression and predictive-forest SHAP percentage contributions by criterion

Variable	Regression			Predictive-forest SHAP			
	CI (%)	CIg (%)	CIc (%)	CI (%)	CIg (%)	CIc (%)	L (%)
Mother's education	23.058	28.726	28.726	16.335	16.335	16.335	17.641
Father's occupation:	17.863	19.885	19.885	12.459	12.459	12.459	11.592
Agriculture							
Mother's occupation:	11.684	14.393	14.393	14.220	14.220	14.220	13.832
Agriculture							
Household wealth	9.116	8.788	8.788	13.727	13.727	13.727	9.019
Type of residence	8.733	11.866	11.866	14.806	14.806	14.806	14.450
Skilled birth attendance	7.848	9.988	9.988	5.446	5.446	5.446	3.247
Region: Equateur	3.761	1.066	1.066	1.697	1.697	1.697	1.603
Father's occupation:	3.370	0.955	0.955	-1.174	-1.174	-1.174	-0.558
Household, unskilled, not working							
Region: Kinshasa	3.221	0.647	0.647	10.330	10.330	10.330	13.100
Region: Sud-Kivu	2.006	0.213	0.213	-0.853	-0.853	-0.853	-0.460
Region: Kasai Oriental	1.746	0.513	0.513	0.425	0.425	0.425	0.546
Father's education	1.462	0.869	0.869	3.279	3.279	3.279	3.723
Region: Maniema	1.075	0.083	0.083	1.387	1.387	1.387	1.820
Region: Orientale	0.984	0.251	0.251	0.616	0.616	0.616	-0.020
Region: Bas-Congo	0.678	0.060	0.060	-0.131	-0.131	-0.131	-0.494

1.2.3 4.2.3 Root-Node Inequality

Table 3: Root-node inequality in under-five mortality by concentration-index criterion

Root node	CI	CIg	CIc	L
Full sample	0.1156	0.0119	0.0477	0.1935

1.2.4 4.2.4 Selected Concentration-Index Tree

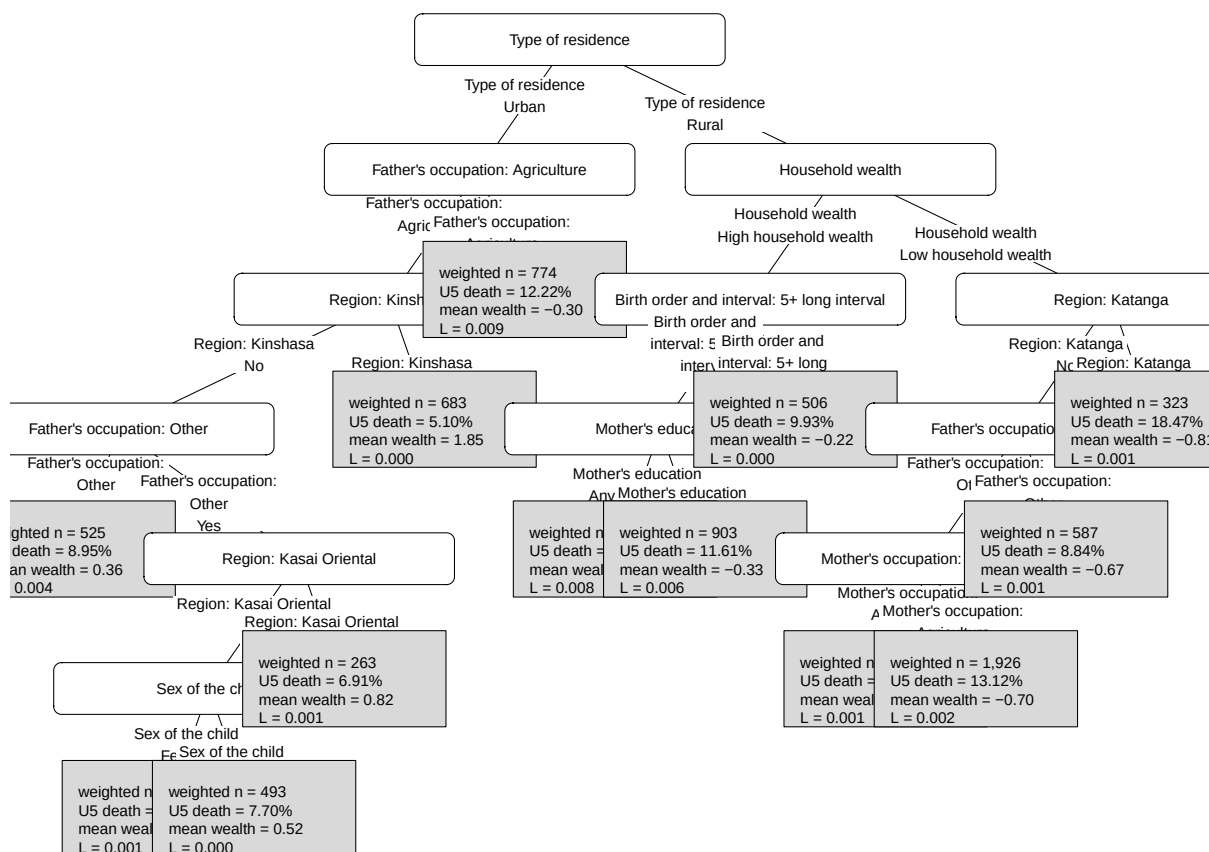


Figure 2: Selected concentration-index tree for under-five mortality in the DRC.

1.2.5 Surrogate Tree Comparison

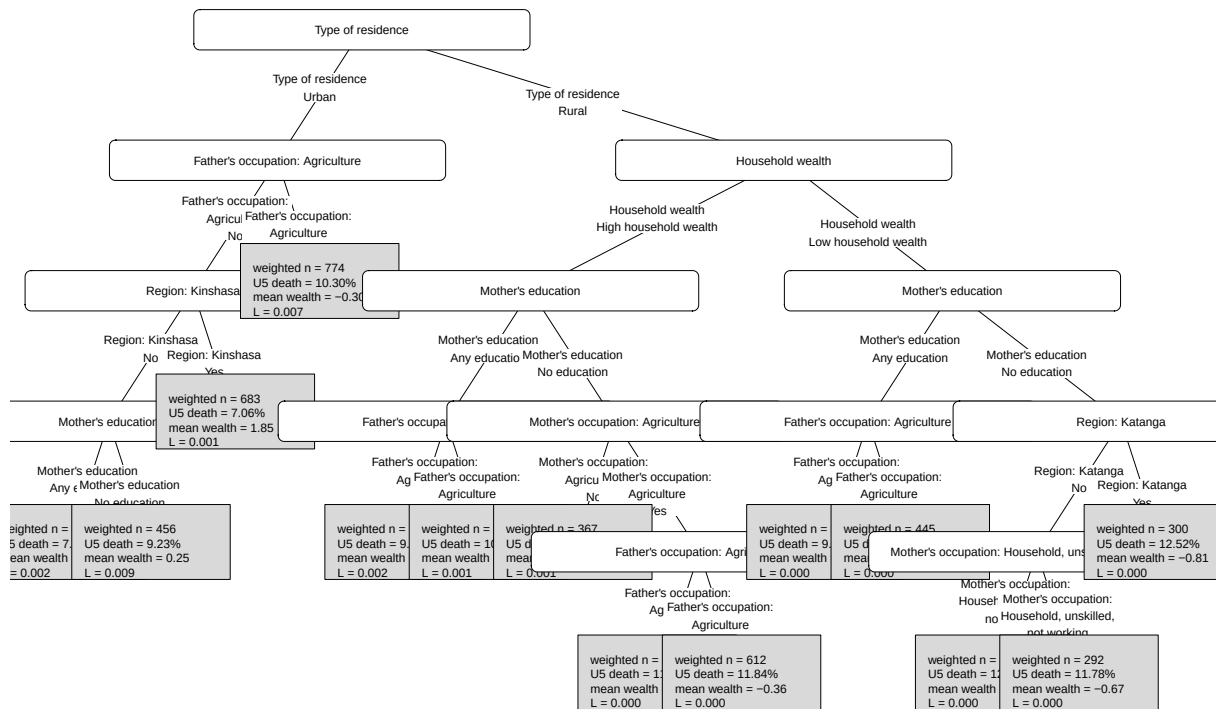


Figure 3: Surrogate tree summarising the selected concentration-index forest predictions.

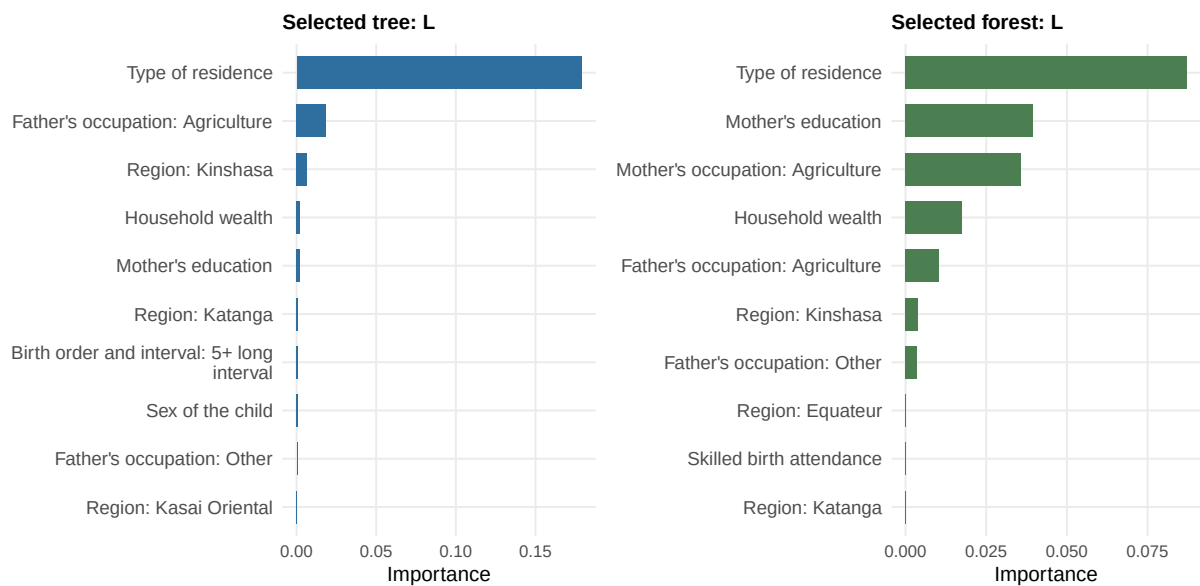


Figure 4: Variable importance for the selected concentration-index tree and forest.

1.3 4.3 Simulation Study

1.3.1 4.3.1 Recovery of the Simulated Subgroup Structure

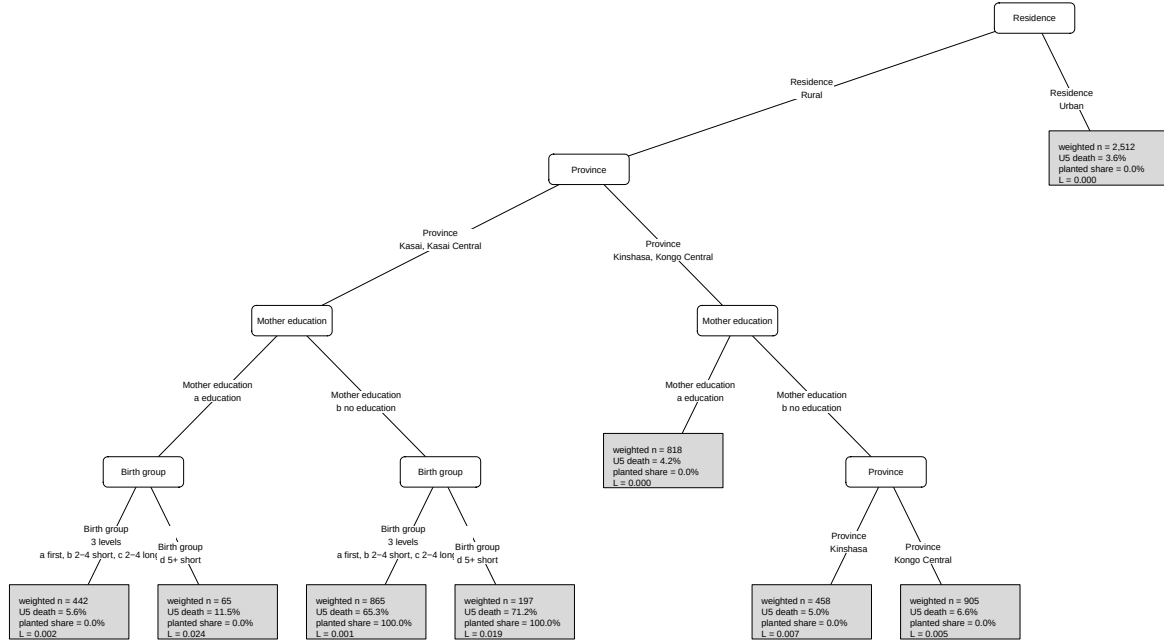


Figure 5: Tree recovered from simulated data with a known subgroup-generating mechanism.

2 Appendix

2.1 Results

2.1.1 Appendix R1: Exploratory Data Outputs

Table 4: Weighted distribution of analysis variables by under-five mortality status

Variable	Under-five mortality		Overall
	Alive at age 5	Died before age 5	Overall
	(weighted N=7,464)	(weighted N=859)	(weighted N=8,323)
Household wealth			
High household wealth	4,266.9 (57.2%)	418.6 (48.7%)	4,685.5 (56.3%)
Low household wealth	3,197.4 (42.8%)	440.2 (51.3%)	3,637.6 (43.7%)
Skilled.birth.attendance			
Skilled.birth.attendance.1	3,225.3 (43.2%)	295.1 (34.4%)	3,520.5 (42.3%)
No.skilled.birth.attendance	4,239 (56.8%)	563.6 (65.6%)	4,802.6 (57.7%)
Sex of the child			
Female	3,828.8 (51.3%)	390.4 (45.5%)	4,219.2 (50.7%)
Male	3,635.5 (48.7%)	468.4 (54.5%)	4,103.9 (49.3%)
Birth order and interval			
First birth	1,396.8 (18.7%)	190.9 (22.2%)	1,587.7 (19.1%)
2-4 short interval	774.3 (10.4%)	110.1 (12.8%)	884.4 (10.6%)

2-4 long interval	2,603.8 (34.9%)	216.6 (25.2%)	2,820.3 (33.9%)
5+ short interval	696.6 (9.3%)	150.6 (17.5%)	847.2 (10.2%)
5+ long interval	1,992.8 (26.7%)	190.7 (22.2%)	2,183.5 (26.2%)
Mother's age at birth			
20 or more	4,256.1 (57.0%)	472.6 (55.0%)	4,728.8 (56.8%)
Less than 20	3,208.1 (43.0%)	386.2 (45.0%)	3,594.3 (43.2%)
Type of residence			
Urban	2,932.1 (39.3%)	263.6 (30.7%)	3,195.7 (38.4%)
Rural	4,532.2 (60.7%)	595.2 (69.3%)	5,127.4 (61.6%)
Mother's education			
Any education	3,342.9 (44.8%)	278.8 (32.5%)	3,621.8 (43.5%)
No education	4,121.4 (55.2%)	579.9 (67.5%)	4,701.3 (56.5%)
Father's education			
Any education	5,490 (73.6%)	589.1 (68.6%)	6,079.2 (73.0%)
No education	1,974.3 (26.4%)	269.7 (31.4%)	2,243.9 (27.0%)
Mother's occupation			
Other	1,610.2 (21.6%)	135.1 (15.7%)	1,745.4 (21.0%)
Household, unskilled, not working	1,757.2 (23.5%)	172.1 (20.0%)	1,929.4 (23.2%)
Agriculture	4,096.8 (54.9%)	551.6 (64.2%)	4,648.4 (55.8%)
Father's occupation			
Other	2,810.3 (37.6%)	242.1 (28.2%)	3,052.4 (36.7%)
Household, unskilled, not working	961.3 (12.9%)	108.5 (12.6%)	1,069.8 (12.9%)
Agriculture	3,692.7 (49.5%)	508.1 (59.2%)	4,200.9 (50.5%)

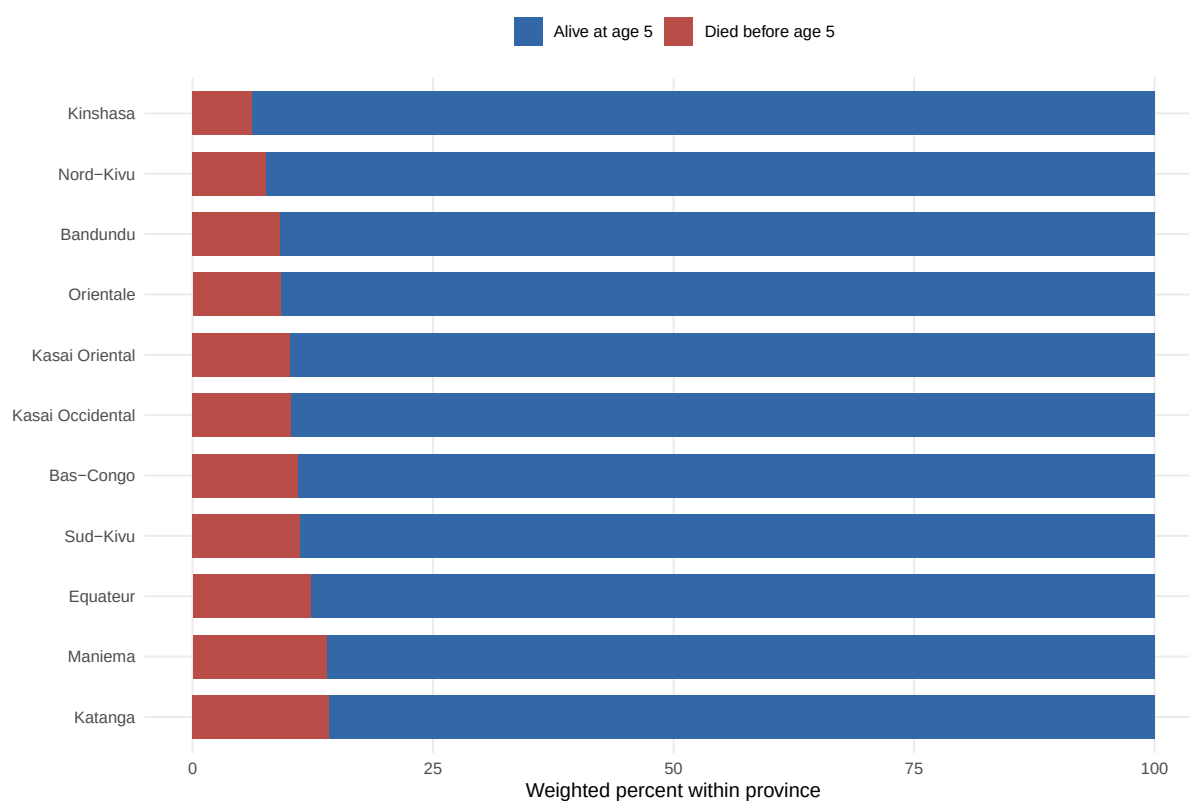


Figure 6: Weighted province distribution by under-five mortality status.

2.1.2 Appendix R2: Regression and Predictive-Forest Details

Regression model summary

Table 5: Survey-weighted regression model summary

Term	Estimate	Std. error	Statistic	p-value	Conf. low	Conf. high
Intercept	-2.908	0.275	-10.563	<0.001	-3.448	-2.369
Household wealth	0.068	0.125	0.544	0.5865	-0.177	0.314
Skilled birth attendance	0.198	0.111	1.783	0.0746	-0.020	0.416
Sex of the child	0.242	0.105	2.294	0.0218	0.035	0.449
Birth order and interval: 2-4 short interval	-0.004	0.186	-0.020	0.9838	-0.367	0.360
Birth order and interval: 2-4 long interval	-0.509	0.169	-3.013	0.0026	-0.840	-0.178
Birth order and interval: 5+ short interval	0.420	0.210	1.995	0.0461	0.007	0.832
Birth order and interval: 5+ long interval	-0.396	0.211	-1.882	0.0599	-0.809	0.017
Mother's age at birth	0.033	0.143	0.229	0.8186	-0.247	0.312
Type of residence	0.090	0.154	0.584	0.5591	-0.212	0.392
Mother's education	0.336	0.132	2.555	0.0106	0.078	0.594
Father's education	-0.036	0.128	-0.280	0.7798	-0.286	0.214
Mother's occupation: Household, unskilled, not working	-0.018	0.155	-0.114	0.9093	-0.322	0.287
Mother's occupation: Agriculture	0.140	0.163	0.861	0.3895	-0.180	0.460
Father's occupation: Household, unskilled, not working	0.235	0.162	1.450	0.1472	-0.083	0.554
Father's occupation: Agriculture	0.225	0.149	1.509	0.1314	-0.067	0.517
Region: Bas-Congo	0.258	0.243	1.062	0.2883	-0.218	0.734
Region: Equateur	0.250	0.238	1.051	0.2934	-0.216	0.715
Region: Kasai Occidental	0.105	0.228	0.460	0.6452	-0.342	0.553
Region: Kasai Oriental	0.153	0.240	0.635	0.5251	-0.318	0.624
Region: Katanga	0.515	0.228	2.258	0.0240	0.068	0.962
Region: Kinshasa	0.081	0.262	0.310	0.7564	-0.433	0.596
Region: Maniema	0.489	0.224	2.178	0.0294	0.049	0.929
Region: Nord-Kivu	-0.252	0.294	-0.857	0.3915	-0.827	0.324
Region: Orientale	-0.084	0.265	-0.316	0.7522	-0.603	0.435
Region: Sud-Kivu	0.300	0.270	1.111	0.2667	-0.229	0.829

Predictive ranger tuning results

Table 6: Weighted class distribution before and after ranger undersampling

Sample	Outcome	Rows	Weighted N	Weighted percent
Analysis data	Alive at age 5	7424	7464.3	89.68
Analysis data	Died before age 5	872	858.8	10.32

Table 7: Cross-validation results for the predictive ranger model

Metric	mtry	min_n	Mean	Std. error	Folds	Rank
accuracy	1	10	0.8949	0.0001	10	1
accuracy	1	80	0.8949	0.0001	10	2
accuracy	1	150	0.8949	0.0001	10	3
accuracy	10	80	0.8949	0.0001	10	4
accuracy	10	150	0.8949	0.0001	10	5
accuracy	19	80	0.8949	0.0001	10	6
accuracy	19	150	0.8949	0.0001	10	7
accuracy	29	80	0.8949	0.0001	10	8
accuracy	29	150	0.8949	0.0001	10	9
accuracy	10	10	0.8904	0.0011	10	10
accuracy	19	10	0.8860	0.0014	10	11
accuracy	29	10	0.8831	0.0017	10	12
roc_auc	1	10	0.6181	0.0091	10	1
roc_auc	1	80	0.6158	0.0086	10	2
roc_auc	1	150	0.6143	0.0090	10	3
roc_auc	10	150	0.6072	0.0070	10	4
roc_auc	10	80	0.6030	0.0069	10	5
roc_auc	19	150	0.5997	0.0074	10	6
roc_auc	19	80	0.5970	0.0068	10	7
roc_auc	29	150	0.5966	0.0082	10	8
roc_auc	10	10	0.5962	0.0081	10	9
roc_auc	29	80	0.5942	0.0074	10	10
roc_auc	19	10	0.5915	0.0079	10	11
roc_auc	29	10	0.5873	0.0077	10	12
sensitivity	19	10	0.0253	0.0048	10	1
sensitivity	29	10	0.0241	0.0050	10	2
sensitivity	10	10	0.0149	0.0034	10	3
sensitivity	1	10	0.0000	0.0000	10	4
sensitivity	1	80	0.0000	0.0000	10	5
sensitivity	1	150	0.0000	0.0000	10	6
sensitivity	10	80	0.0000	0.0000	10	7
sensitivity	10	150	0.0000	0.0000	10	8
sensitivity	19	80	0.0000	0.0000	10	9
sensitivity	19	150	0.0000	0.0000	10	10
sensitivity	29	80	0.0000	0.0000	10	11
sensitivity	29	150	0.0000	0.0000	10	12

Predictive ranger SHAP summary plot



Figure 7: SHAP summary plot for the predictive ranger model.

2.1.3 Appendix R3: Concentration-Index Tree Details

Tree tuning and selection details

Table 8: Best tree hyperparameters selected by validation gain within each impurity criterion

type	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	500	250	0.01	10	0	0.2
CIg	500	250	0.01	10	0	0.2
CIc	500	250	0.01	10	0	0.2
L	500	250	0.01	10	0	0.1

Tree fit and selection summaries

Table 9: Tree summary for the selected settings

crit	metric	estimate	std_error
CI	Root impurity	0.1191	NA
CI	Training gain	0.1000	0.0022
CI	Training gain (% root)	86.6637	1.2283
CI	Validation gain	-0.0399	0.0311
CI	Validation gain (% root)	-111.8420	54.0757
CIc	Root impurity	0.0483	NA
CIc	Training gain	0.0414	0.0008
CIc	Training gain (% root)	86.8643	1.1416
CIc	Validation gain	-0.0137	0.0140
CIc	Validation gain (% root)	-85.3460	40.9027
CIg	Root impurity	0.0121	NA
CIg	Training gain	0.0103	0.0002
CIg	Training gain (% root)	86.8643	1.1416
CIg	Validation gain	-0.0034	0.0035
CIg	Validation gain (% root)	-85.3460	40.9027
L	Root impurity	0.3128	NA
L	Training gain	0.1912	0.0062
L	Training gain (% root)	98.4497	0.0797
L	Validation gain	0.2797	0.1070
L	Validation gain (% root)	63.9740	16.6828

Tree selection metrics

Table 10: Selection metrics for fitted concentration-index trees

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Terminal nodes
CI	0.1191	0.1000	0.8666	-0.0399	-1.1184	7.3
CIc	0.0483	0.0414	0.8686	-0.0137	-0.8535	7.6
CIg	0.0121	0.0103	0.8686	-0.0034	-0.8535	7.6
L	0.3128	0.1912	0.9845	0.2797	0.6397	9.1

Tree validation plots

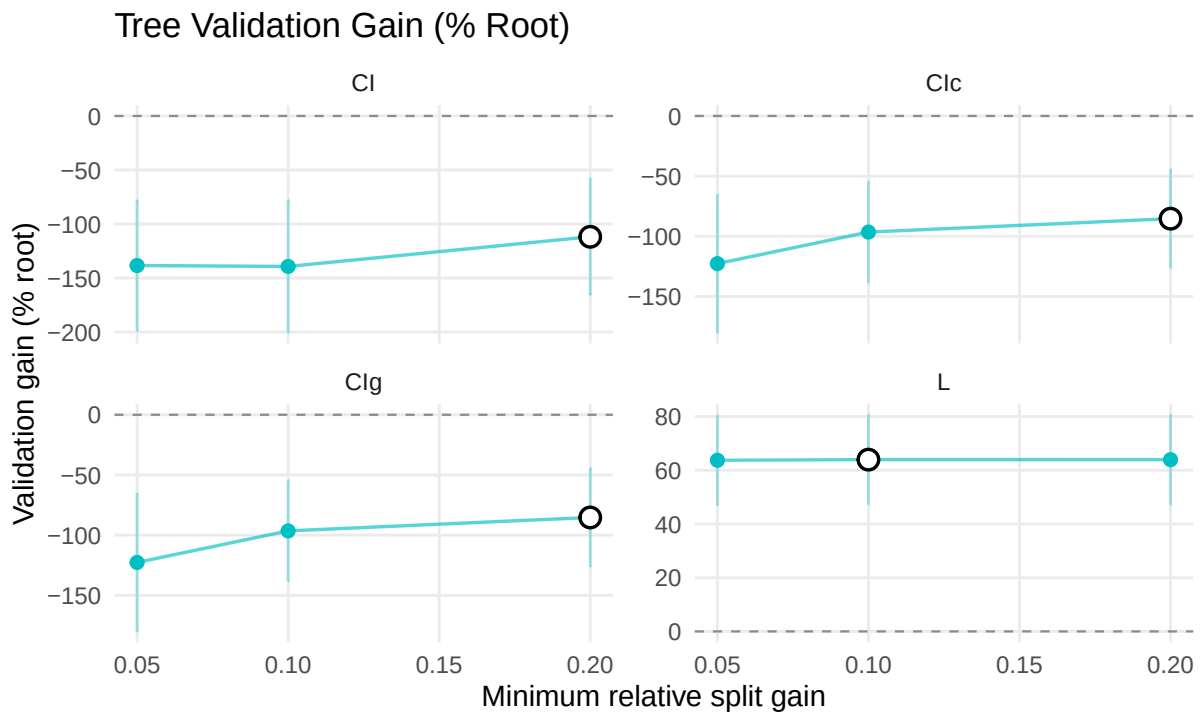


Figure 8: Tree validation gain and complexity by minimum relative split gain.

Tree complexity plots

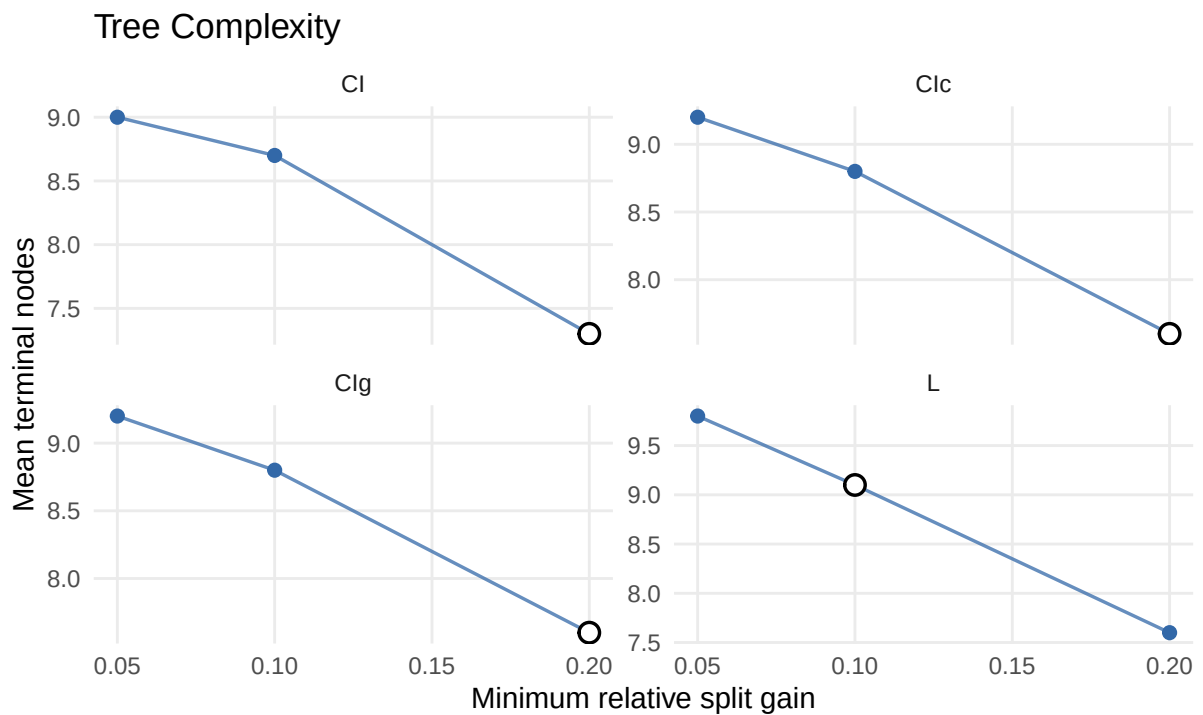


Figure 9: Tree complexity by number of terminal nodes.

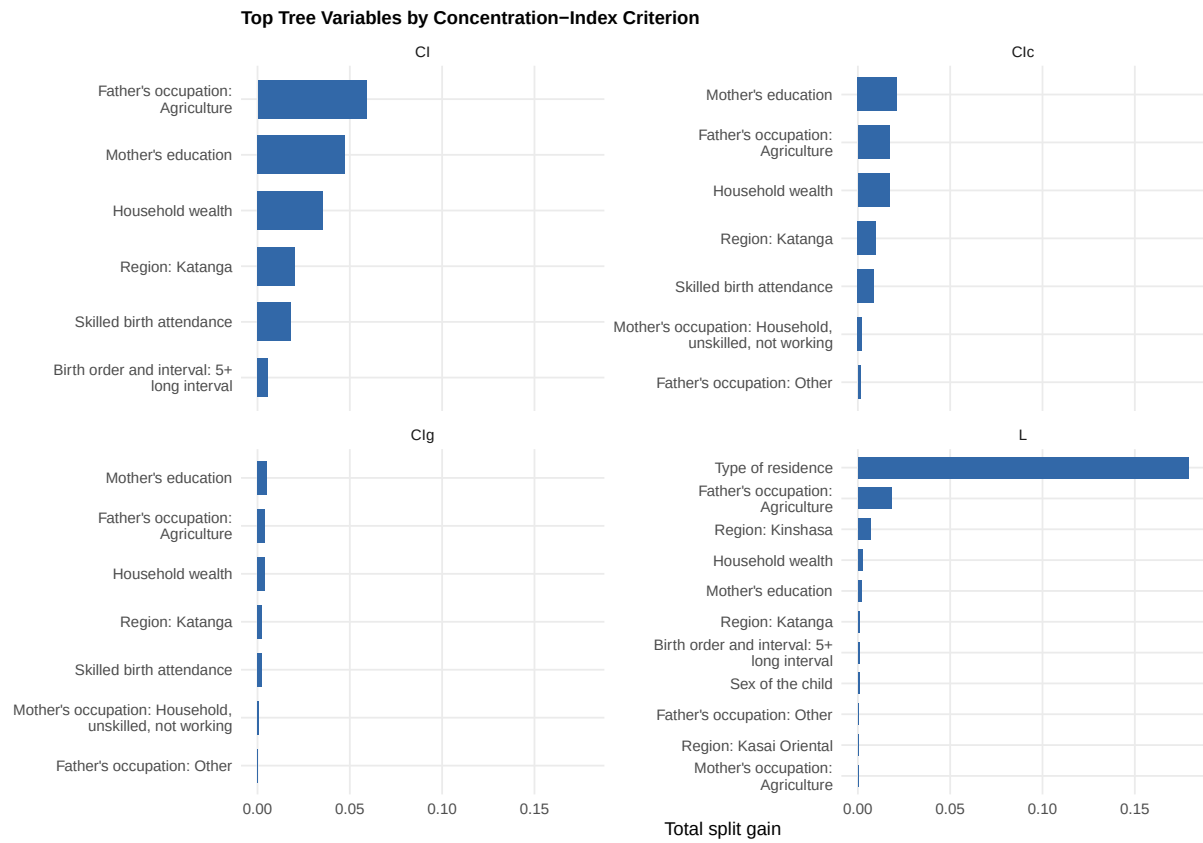


Figure 10: Top tree variables by concentration-index criterion.

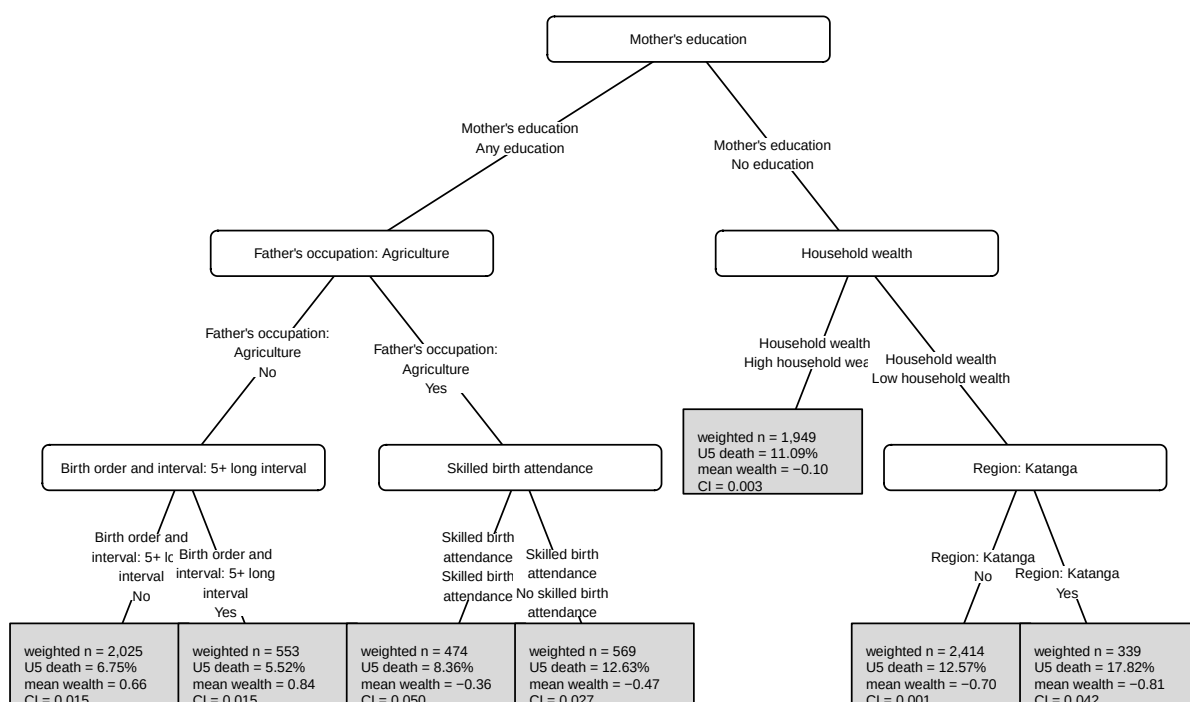


Figure 11: Concentration-index tree fitted with the CI criterion.

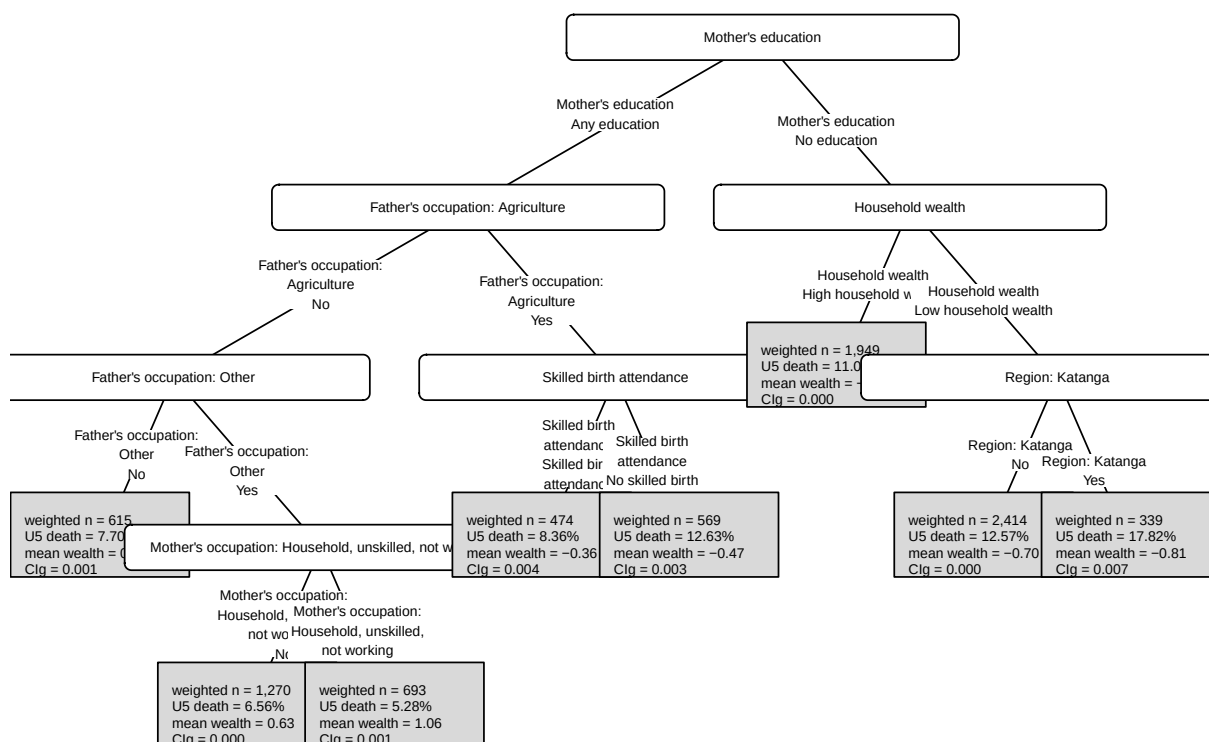


Figure 12: Concentration-index tree fitted with the CIg criterion.

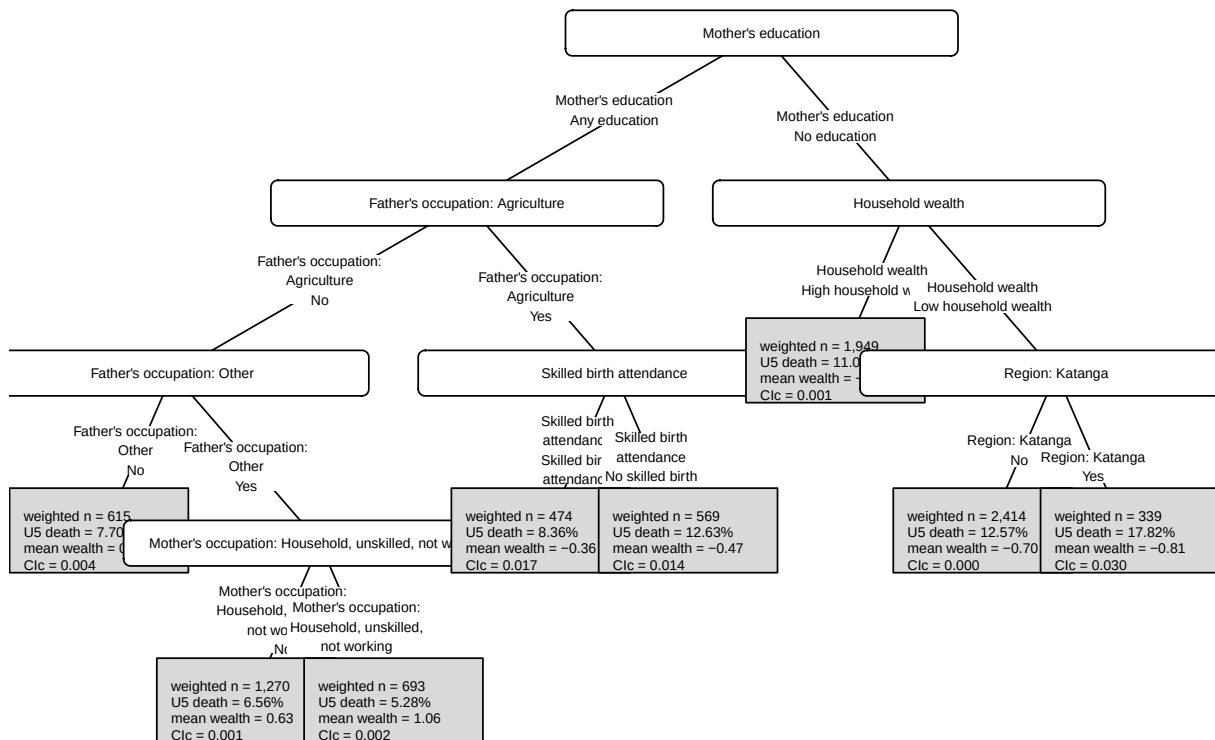


Figure 13: Concentration-index tree fitted with the CIc criterion.

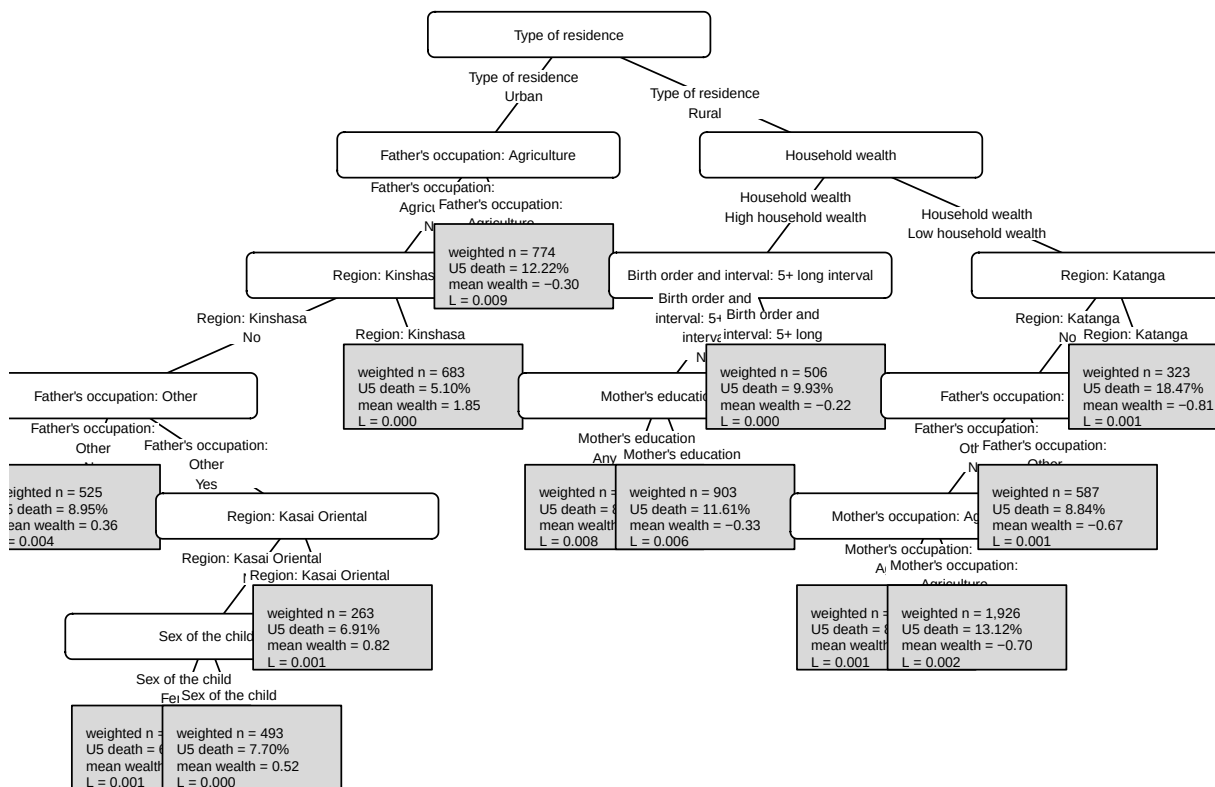


Figure 14: Concentration-index tree fitted with the L criterion.

Lecturer rpart comparison trees

The lecturer-style `rpart` fits are converted to `party` trees before plotting, so the figures use the same terminal-node summaries as the concentration-index tree plots.

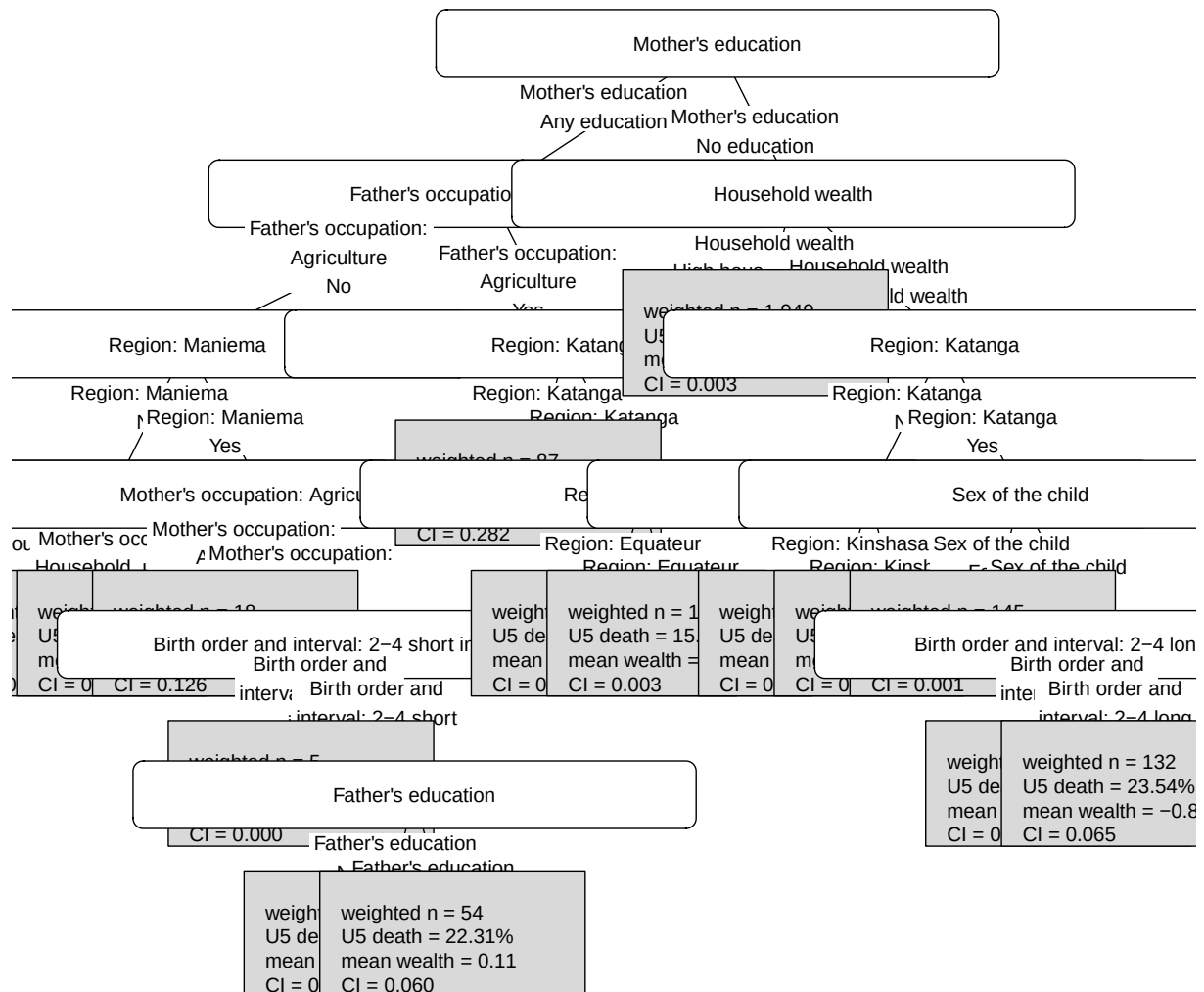


Figure 15: Lecturer rpart comparison tree using absolute CI impurity.

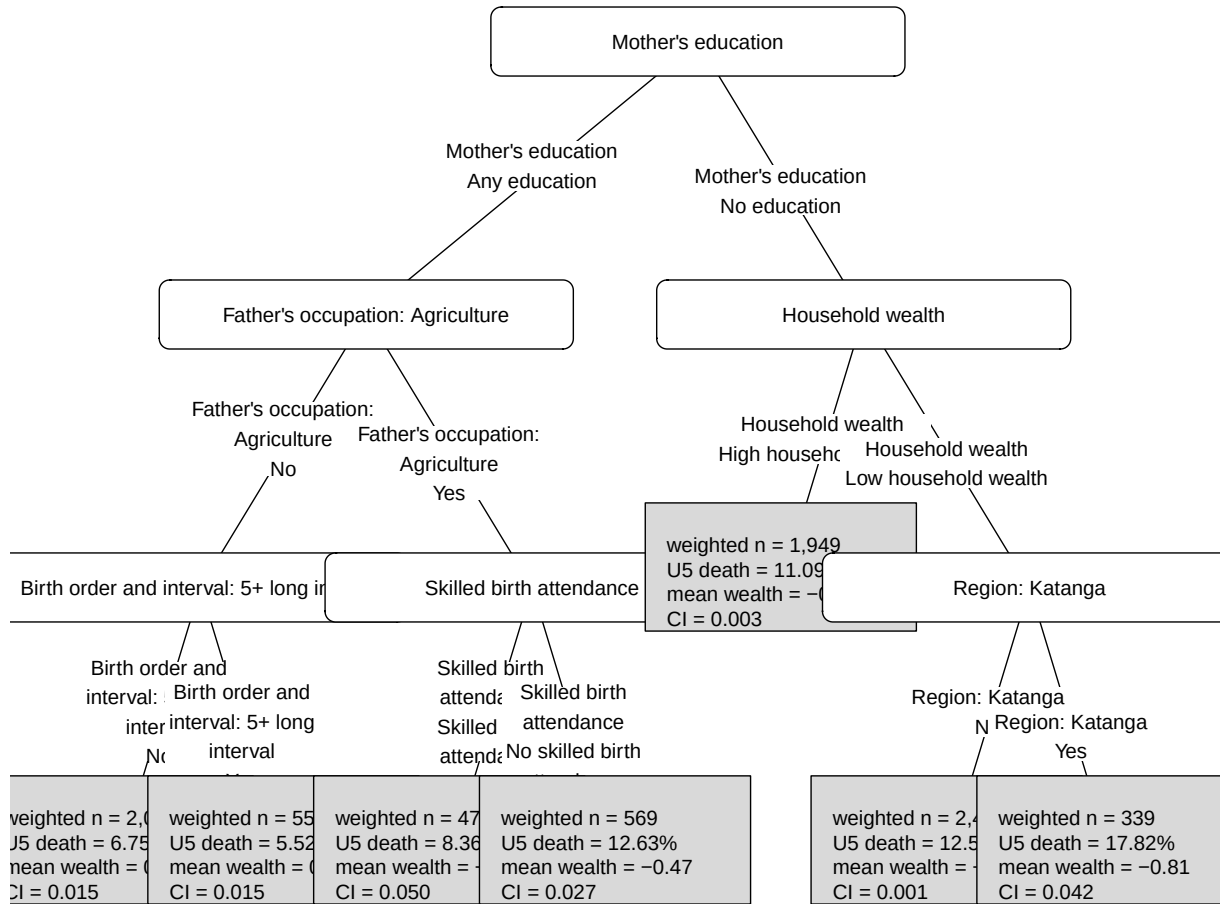


Figure 16: Lecturer rpart comparison tree using robust absolute CI impurity.

2.1.4 Appendix R4: Concentration-Index Forest Details

Forest tuning and selection details

Table 11: Best forest hyperparameters selected by validation gain within each impurity criterion

type	ntree	mtry	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	100	5	500	250	0.01	10	0	0.3
CIc	100	5	500	250	0.01	10	0	0.3
CIg	100	5	500	250	0.01	10	0	0.3
L	100	14	500	250	0.01	10	0	0.3

Forest fit and selection summaries

Table 12: Forest summary for the selected settings

criterion	metric	estimate	std_error
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CI	Root impurity	0.1191	NA
CI	Training gain	0.0161	0.0017
CI	Training gain (% root)	14.0579	1.4936
CI	Validation gain	-0.0043	0.0061
CI	Validation gain (% root)	-13.9412	7.4702
CIc	Root impurity	0.0483	NA
CIc	Training gain	0.0092	0.0005
CIc	Training gain (% root)	19.4678	1.0964
CIc	Validation gain	-0.0014	0.0030
CIc	Validation gain (% root)	-17.8481	10.7955
CIg	Root impurity	0.0121	NA
CIg	Training gain	0.0024	0.0002
CIg	Training gain (% root)	19.9279	1.5373
CIg	Validation gain	-0.0004	0.0008
CIg	Validation gain (% root)	-17.6840	10.2521
L	Root impurity	0.3128	NA
L	Training gain	0.1837	0.0062
L	Training gain (% root)	94.5564	0.1551
L	Validation gain	0.2771	0.1068
L	Validation gain (% root)	69.2622	12.6936

Forest validation plots

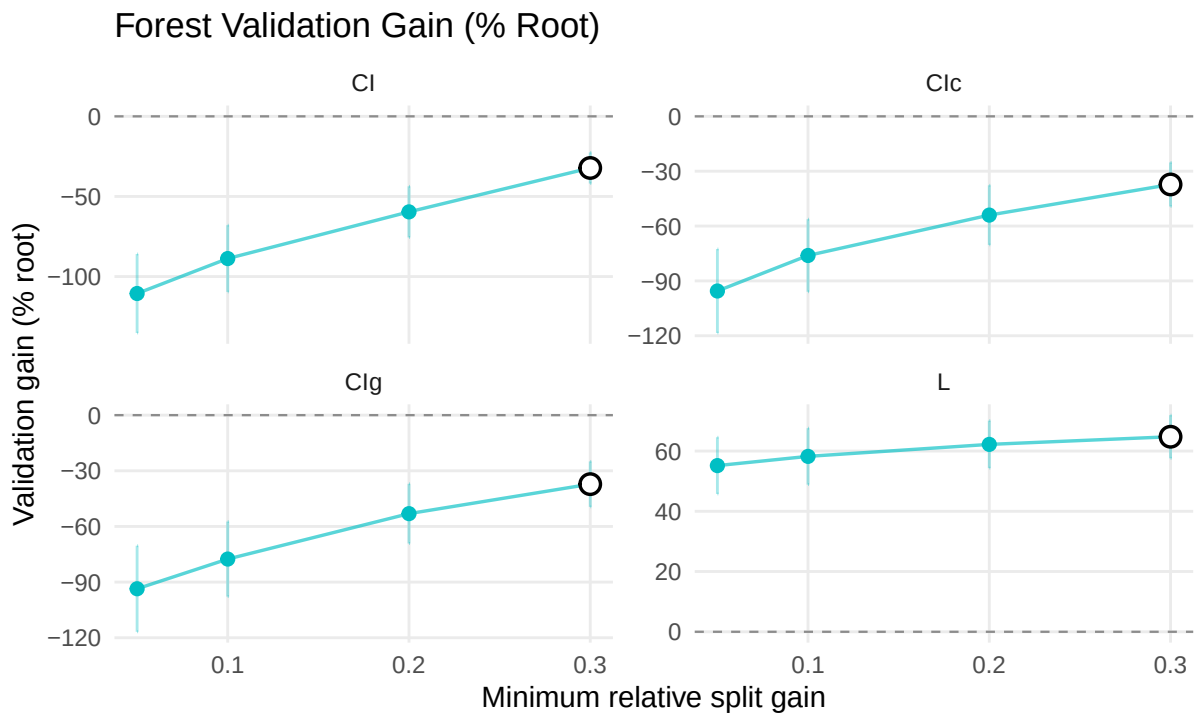


Figure 17: Forest validation gain and complexity by minimum relative split gain.

Forest complexity plots

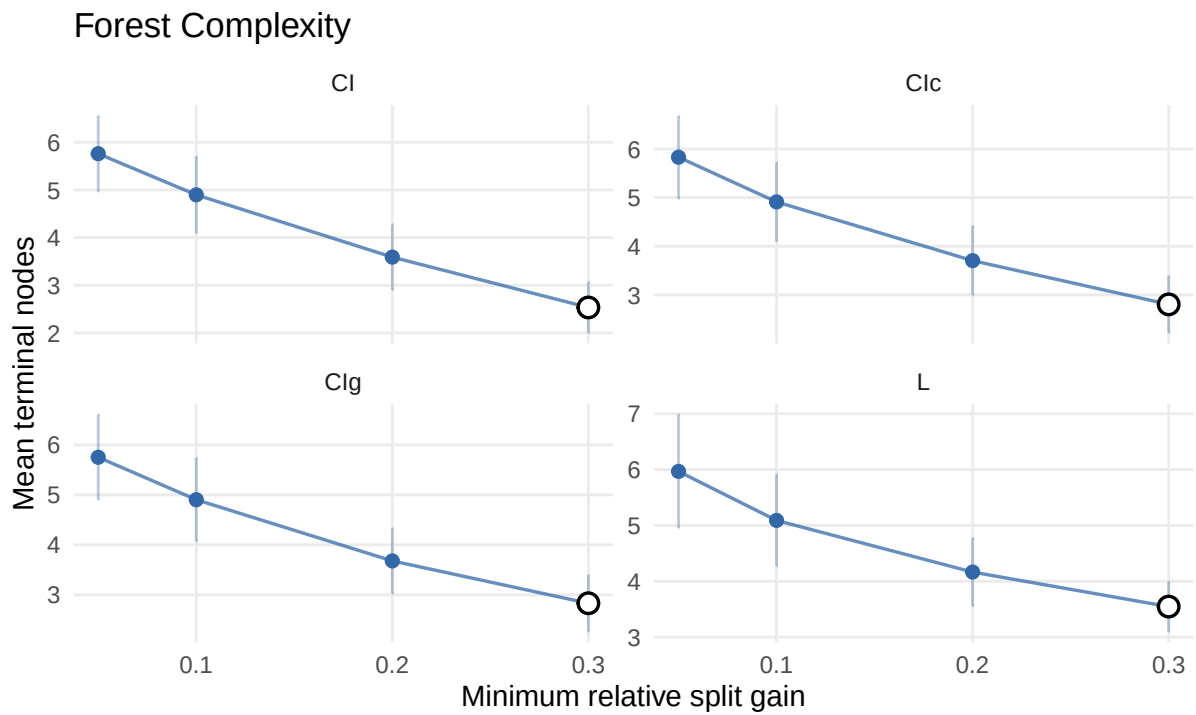


Figure 18: Forest complexity by number of terminal nodes.

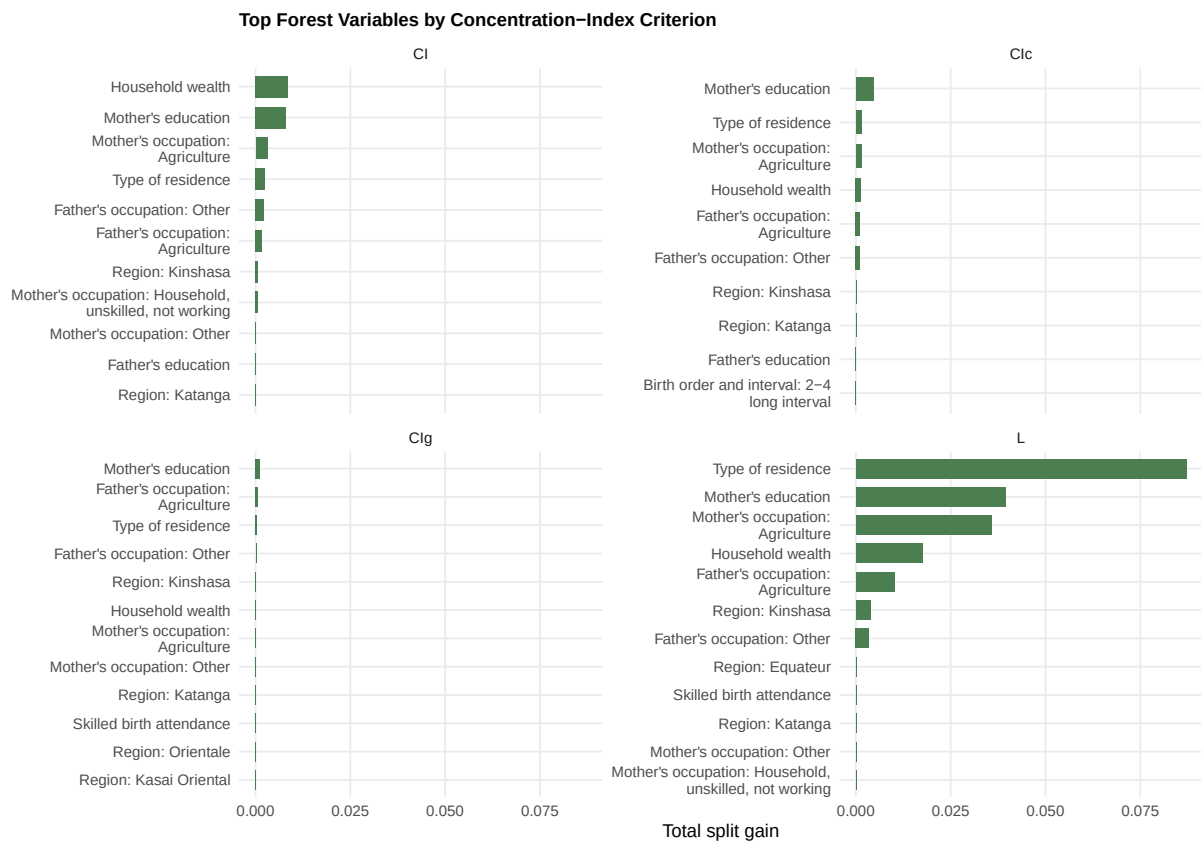


Figure 19: Top forest variables by concentration-index criterion.

Non-Selected Forest Surrogate Trees

The selected forest surrogate appears in the results section. The remaining criterion-specific forest surrogates are retained here for comparison.

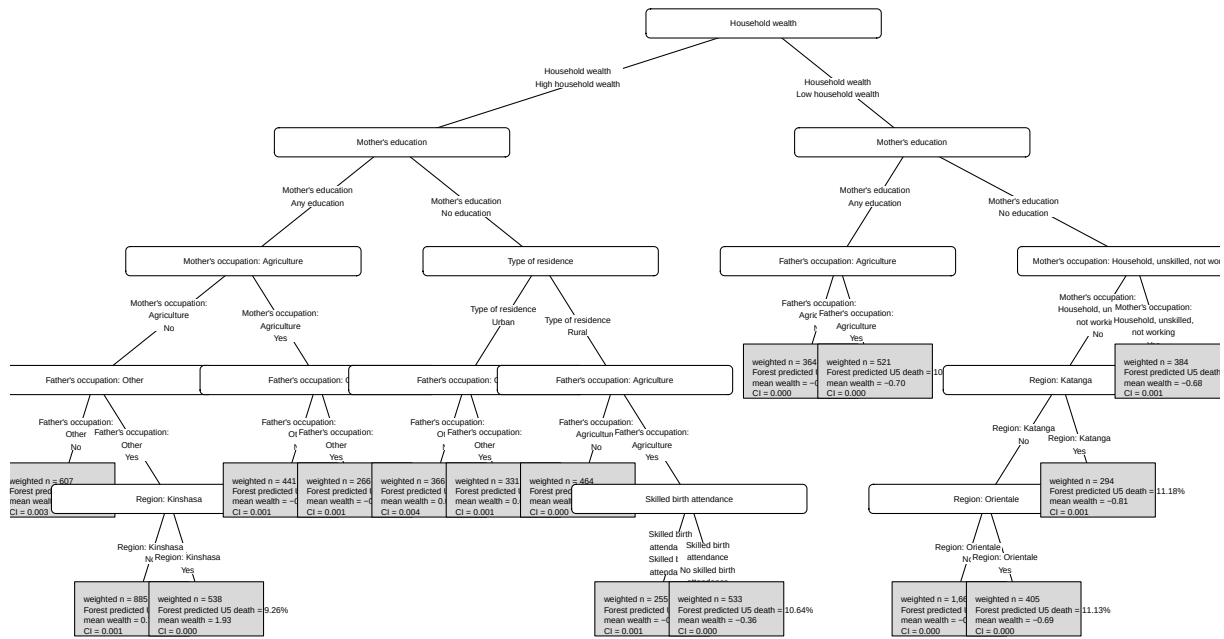


Figure 20: Surrogate tree for the CI concentration-index forest.

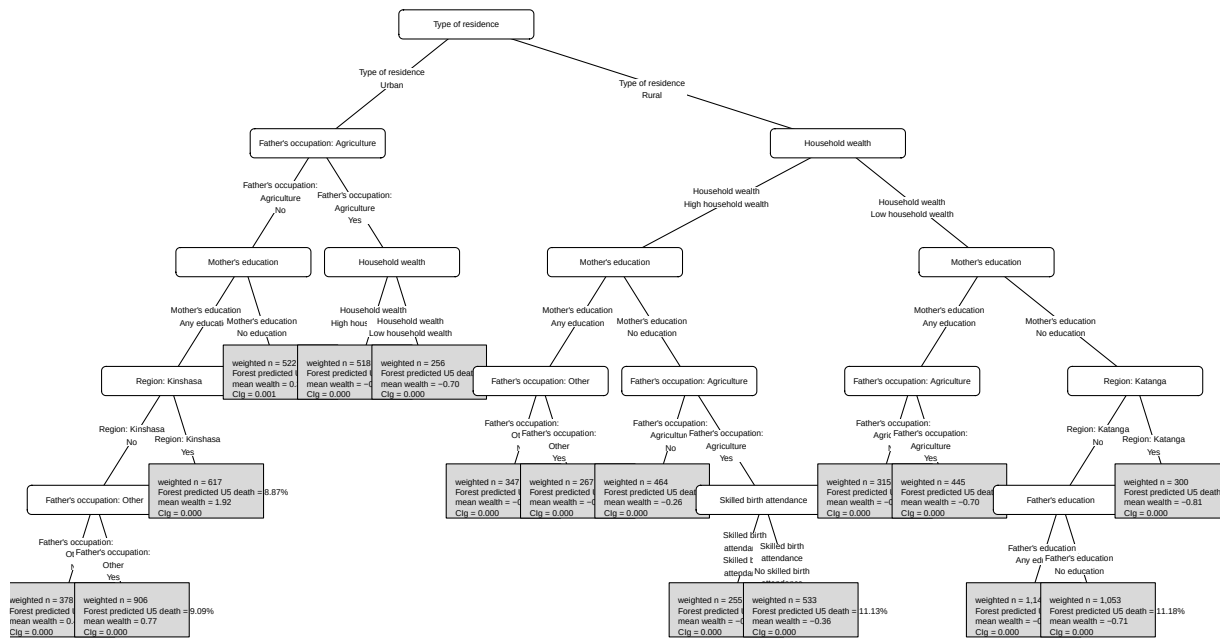
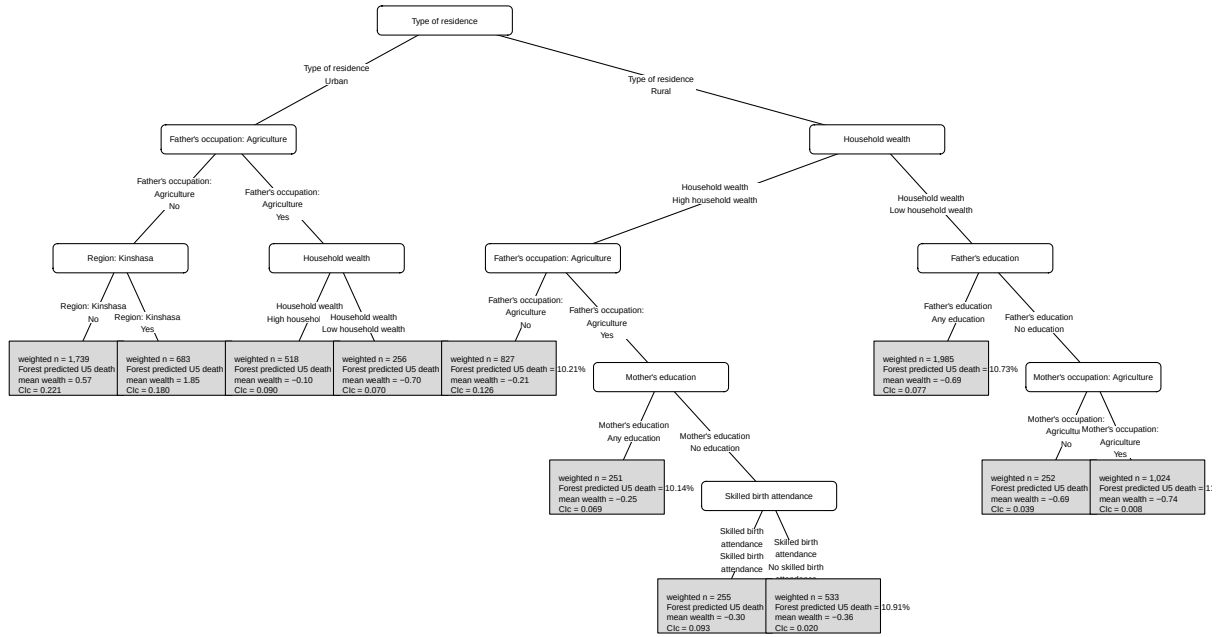


Figure 21: Surrogate tree for the CIg concentration-index forest.



2.1.5 Appendix R5: Implementation Status

Implemented in the report: selected concentration-index tree and selected forest/surrogate-tree variable-importance plots in the results section; tree and concentration-index forest best-control tables, selected-setting validation summaries, selection metrics, validation and complexity plots, and variable-importance plots by criterion in the appendix; ranger cross-validation and SHAP summary outputs in the appendix.

To implement for the final analysis run: rerun the generator with `RUN_CV=true` and `RUN_FOREST_CV=true` when the full cross-validation objects should be recomputed from scratch rather than read from previously generated results.